DEPARTMENT OF MATHEMATICAL SCIENCES
(Faculty of Science)
SA'ADU ZUNGUR UNIVERSITY BAUCHI
End of Second Semester Examination 2024/2025 Session

Course Code and Title: CSC2308-FOUNDATION OF SEQUENTIAL PROGRAMME **Time Allowed: (2 Hours)**

Credit Units: (3 Units)
Instruction: (Answer any three (3) questions)

Q1)

a) Highlight any three (3) roles that the Operating System plays in relationship between I.T and Computer Architecture. 3marks

b) Draw the diagram of a Language Processing System. 3marks

c) Explain any three (3) of the following terms in compiler: 3marks

i) Scanning
ii) Parsing
iii) Semantic Analysis
iv) Intermediate Code Generation
v) Code Optimization

Q2)

a) Use a diagram to describe the Fetch, Decode, and Execute life cycle in Von Neumann architecture. 5marks

b) Use IEEE 754 single precision floating point number representation to find the decimal equivalence of: 1 10011011 01 10000 0000 0000 0000 0000 5marks

c) Use 4-bit Signed Integer Representation to perform the following operations: 5marks

i) 0 + (-2) in Sign Magnitude
ii) 3 + (-3) in 1's Complement
iii) (-4) + (-1) in 1's Complement
iv) (-4) + (-1) in 2's Complement

Q3)

a) What do you understand by the term Instruction Selection? 3marks

b) Given the following equation: $z = a^2 + 4y + 8$

i) Generate an Abstract Syntax Tree (AST) for the equation. 3marks

ii) Use the following latencies for the instructions/operations: load = 2cycles, mul = 3cycles, div = 5cycles, store = 2cycles and sub = 1cycle to find the total latency required to schedule the instructions. (Assuming a single issue machine) (Hint: P = 1) 4marks

iii) Perform Instruction scheduling to minimized the total latency and show the peak register pressure (PRP). 3marks

c. Perform a Register Allocation to the generated AST, assuming that the machine has only two (2) registers P0 & P1. 2marks

Q4)

a) i) Define a Regular Language in terms of a Regular Grammar. 3marks

ii) Convert 0 1101 to Binary. 2marks

iii) Convert 7 0611 1119 4011 0111 0H1 to HEX. 2marks

GOOD LUCK
